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HAMMER-V4 Linux-Booted RV64IMA SoC Subsystem

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 1 November 2026 (2026-11-01 09:46 UTC+8 Beijing Time)

Abstract

HAMMER-V4 extends the preserved HAMMER-V3 platform with a synthesizable boot-ROM loader and a reproducible Linux 6.6.1 software stack. The single-hart subsystem supports RV64IMA, M/S/U privilege, Sv39 translation, coherent memory access, CLINT, PLIC and a 16550A console. Software-vs-Hardware numerical comparison used an independent Python architectural and timing oracle against compiled RTL execution. Seeds 11, 29 and 47 each matched all 3,600 UART bytes, including Linux startup, U-mode /init and the expected PID-1 exit panic. A separate clean-source seed-11 reproduction also matched. These results establish simulated RTL parity, not on-board Linux execution. [E223] Native L1 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA achieved WNS_setup = +0.278 ns and hold slack = +0.039 ns at 100 MHz. This standalone implementation uses 1,967 virtual pins and zero physical pins; all non-clock input paths and all output paths are false-pathed. The result establishes internal clock-domain timing, not external interface or board-level timing closure. L3 physical Arria 10 HIL passed three deterministic hardware-bridge connectivity probes, not the Linux workload. [E225–E227]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

All 40 inherited RTL files are preserved, and hammer_v4_top literally instantiates hammer_v3_top as u_v3 inside its legal-configuration wrapper. Compatibility takes precedence over replacing the reset architecture: a registered loader installs a 32-word trampoline through the existing halted host protocol, preserving cache-maintenance ordering. Firmware, Linux and the DTB remain addressed images in external RAM. The processor retains variable-latency memory paths, separate instruction/data TLBs, a shared page-table walker, direct-mapped caches and sequential multiply/divide. Blocks are arranged row-major in three columns; the software block represents executed instructions rather than additional hardware. [E001, E082–E098]

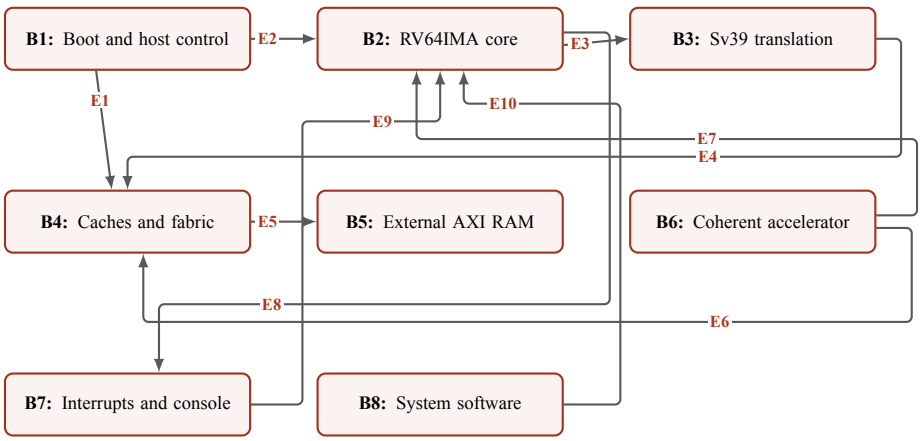


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Boot and host control	Arbitrates external host traffic against the ROM loader, gates launch and accelerator admission, and exposes boot completion or reset-only failure.
B2: RV64IMA core	Executes M/S/U instructions with pipeline interlocks, prediction, precise traps, privileged CSRs, atomics and iterative arithmetic.
B3: Sv39 translation	Separate ITLB/DTLB ports and a shared walker enforce canonicality, permissions, superpage alignment and fence semantics.
B4: Caches and fabric	Physically tagged instruction/data caches, on-chip backing memories and registered arbitration preserve ownership and maintenance ordering.
B5: External AXI RAM	Platform-provided RAM contains firmware, Linux, page tables, user memory and the reserved DTB.
B6: Coherent accelerator	The AXI4 slave bridge drains the core and performs cache maintenance before exclusive backing-memory transactions.
B7: Interrupts and console	CLINT supplies machine time/software interrupts; PLIC routes interrupt sources; UART provides serialized RX/TX.
B8: System software	Original OpenSBI-compatible firmware enters upstream Linux 6.6.1, which runs the embedded static /init under virtual memory.

Directed relationship	Meaning
E1: B1 to B4	Host loading and maintenance
E2: B1 to B2	Gated start
E3: B2 to B3	Virtual instruction/data requests
E4: B3 to B4	Translated requests and PTE reads
E5: B4 to B5	External memory transactions
E6: B6 to B4	Exclusive coherent access
E7: B6 to B2	Drain and resume control
E8: B2 to B7	Peripheral register accesses
E9: B7 to B2	Interrupt delivery
E10: B8 to B2	Executed firmware/kernel/user instructions

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n, start, halt_req; boot_ready, boot_error, halted	Control inputs; status outputs	1 bit each	One 100 MHz clock domain with active-low asynchronous reset. Start before readiness is discarded, not queued. Boot failure prohibits launch until reset.
host_*	Request inputs; readiness and response outputs	32-bit address, 64-bit data, 8-bit write strobes; scalar controls	Inherited halted instruction/data loading interface. Internal boot traffic is exclusive: external host_ready stays low while loading and internal responses remain hidden. Simultaneous ready-state host traffic and start retain host priority.

Signal / group	Direction	Width	Contract
m_mem_axi_*	AXI4 memory master	64-bit address/data, 4-bit ID, 8-bit length/ strobes, 2-bit responses	Backing requests become aligned single-beat 64-bit INCR transfers with ID 0, size 3 and length 0. AW/W accept independently; payloads remain stable under backpressure. Bad IDs, response errors or missing RLAST produce errors.
s_acc_axi_*	Coherent AXI4 slave	64-bit address/data, 4-bit ID, 8-bit length/strobes	Retains burst geometry, strobes, response backpressure and exclusive maintenance ordering. Request admission is blocked until wrapper readiness.
m_axi_*	AXI-Lite peripheral master	32-bit address, 64-bit data, 8-bit strobes, 2-bit responses	Preserves the parent's external peripheral interface without changing channel names or directions.
uart_rx, uart_tx; plic_ sources, irq_meip, irq_ seip	Serial/ interrupt inputs; serial output	1-bit RX/TX and direct IRQs; 31-bit PLIC source vector	The delivered workload uses 6,250,000 baud, 8N1, LSB-first and idle-high signaling. UART is PLIC source 1; vector bits map to sources 2–32.
dbg_*, retire_*, trap_*, stall_*, redirect_*, ex_*	Debug selector input; observation outputs	64-bit PCs/ data/causes; 32-bit instruction; 5-bit register indices; 2-bit privilege/ selectors; scalar flags	Architectural and pipeline observations distinguish executed instructions, memory operations and traps from console claims. Retirement observations are registered.

Parameter	Default	Meaning / supported values
IMEM_WORDS	16384	1–16384; boot mode requires at least 32.
DMEM_WORDS	8192	1–8192; compatibility mode retains singleton and non-power-of-two depths.
PRED_MODE	1	0: static prediction; 1: tagged BTB prediction.
BTB_ENTRIES	16	1, 4, 16 or 64.
ICACHE_LINES, DCACHE_LINES	64 each	1, 4, 16 or 64 each; 32-byte lines.
ITLB_ENTRIES, DTLB_ ENTRIES	8 each	1, 4, 8 or 16 each.
LEGACY_TRAPS	0	0 or 1; boot mode requires 0.
BOOT_ENABLE	1	0 or 1. Zero bypasses ROM installation. Illegal tuples emit PARAMETER ERROR rather than silently clamping.

III Behavior and corner cases

A Boot sequencing

Reset clears loader state and boot flags, suppresses new requests and launch, and leaves the parent halted with UART idle high. The first released edge enters ISSUE. Each accepted ROM write enters WAIT_RESPONSE; only its successful response advances the index. The final response sets boot_ready. Compatibility mode becomes ready on its first released edge. Requests sample pre-edge values; state changes appear post-edge. The trampoline establishes a0=0 and a1=0x87e00000 before jumping to firmware at 0x80000000.

B Integer and virtual-memory semantics

XLEN arithmetic wraps at 64 bits; word results sign-extend. Sequential division preserves divide-by-zero and signed-overflow behavior. Sv39 uses three 512-entry levels, 4 KiB pages, 44-bit satp PPNs and 16-bit ASIDs, with 2 MiB/1 GiB leaves. Canonicality, SUM/MXR, permissions and alignment apply on translation. Missing A, or D on writes, causes a software-managed page fault; hardware does not update those bits. Scoped fences preserve global entries when an ASID is specified.

C Platform map and interrupts

RAM occupies 128 MiB at 0x80000000. Firmware reserves the first 2 MiB; Linux enters physically at 0x80200000; the DTB reserves the final 2 MiB beginning at 0x87e00000. CLINT is at 0x02000000, PLIC at 0x0c000000 and UART at 0x10010000. CLINT time advances once per released clock, including stalls. PLIC priorities are 0–7 with lowest-ID tie breaking.

D Firmware and Linux execution

The original firmware implements SBI v0.2 base functions 0–6, TIME and legacy console/timer calls; unsupported services return NOT_SUPPORTED. It delegates appropriate traps while retaining supervisor ecalls in M-mode, preserves trap context and converts machine timer expiry into supervisor timer delivery. Linux parses the DTB, enables Sv39 and enters /init through sret. The program checks demand-allocated data, write, uname, getpid and exit, and consumes UART input. Successful exit(0) by PID 1 intentionally produces Linux's genuine init-exit panic.

E Software compatibility and reproducibility

Source-controlled patches restore configurable random-source trust and provide S-mode integer misaligned-access emulation without disabling Linux's unaligned-access probe. The handler uses fault-protected byte accesses, preserves extension/x0 semantics and advances EPC only on success; failed stores may expose earlier bytes. Fixed build identity, initramfs metadata and public test entropy make the console reproducible. Explicit object names and packaged timestamps remove path-dependent image differences.

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison and complete Linux boot

Method. Independent Python RV64IMA/Sv39/platform expectations were frozen before compiled Verilator execution; external UART framing, complete bytes and architectural events were compared.

Result. Seeds 11/29/47 each matched 3,600 bytes. Final edges were 1,075,814,213 / 1,076,072,595 / 1,076,092,272, below the 4,294,967,296-edge bound. Event comparisons numbered 317,076 / 317,152 / 317,155 respectively. This establishes software-oracle versus simulated RTL parity, not physical workload parity.

Evidence: E223, E120, E135

B Architectural boot milestones

Method. Compared privilege transitions, translation operations, demand accesses, syscalls and interrupt events rather than printed labels.

Result. Each seed observed the M-to-S handoff, six Sv39 activations, 157 sfence events, 18,388 U-mode retirements, three demand faults and all four required syscalls. Supervisor timer deliveries were 898/899/899; each seed recorded 16 UART PLIC claims and 16 completions.

Evidence: E223

C Parameters, compatibility and directed correctness

Method. Compact behavior tests, ELF/ISA/FDT audits, arithmetic/MMU/firmware cases and separate parent/V4 compatibility executions.

Result. 57 legal/illegal parameter cases passed, not a Cartesian product. Parent and compatibility-mode V4 each produced 72 retirements, 82 external reads and 24 external writes over 6,480 edges. Audits checked nine images and rejected seven malformed FDT headers. The S-mode handler's 160 host-unit cases and 1,000 comparisons are supplementary, not standalone RTL trap proof.

Evidence: E223, E101, E103, E114, E121

D Defect sensitivity

Method. Disposable compiled mutants underwent functional comparisons with unchanged controls.

Result. Five controls passed and all 11 mutants were rejected, covering DTB handoff, permissions, timer handling, UART corruption, time-CSR offset, delayed interrupt delivery, coherence, MPRV, global fencing and reserved PTEs.

Evidence: E223, E113

E Full-hierarchy synthesis and L1 RTL timing lint

Method. Yosys hierarchy/connectivity/state checks and native timing-hazard analysis of 42 RTL files.

Result. Yosys retained 12,048 generic cells, 809 state cells and 1,082,784 memory bits, with zero latches or combinational divide/modulo cells. Native L1 RTL timing lint passed with no divider/modulo, mixed-assignment, LUT-clock-gating or loop-unroll hazards.

Evidence: E223, E122, E225

F L2 Intel Quartus Prime 17.1 TimeQuest timing

Method. Post-fit STA at 100 MHz on Arria 10 GX 10AX115N2F45E1SG, using a standalone virtual-pin implementation. Constraints false-path all non-clock inputs and all outputs, excluding external I/O timing.

Result. Passed: WNS_setup = +0.278 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 102.86 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E226

G L3 physical Arria 10 HIL

Method. Three consecutive physical PCIe BAR0 hardware-bridge register probes.

Result. 3/3 deterministic passes, each reading register 57 as 178. The receipt explicitly classifies this as bridge-connectivity-only; it does not demonstrate flashed HAMMER-V4 workload execution or physical UART/oracle equivalence.

Evidence: E227

H Clean-directory reproduction

Method. Rebuilt declared software inputs in a clean copied project, compared images, regenerated independent expectations and executed seed 11 on RTL.

Result. Nine images were byte-identical across builds from 162 declared inputs. The separate clean boot matched all 3,600 UART bytes and reproduced the original seed-11 event counts and final edge.

Evidence: E223, E118

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	8 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 52	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

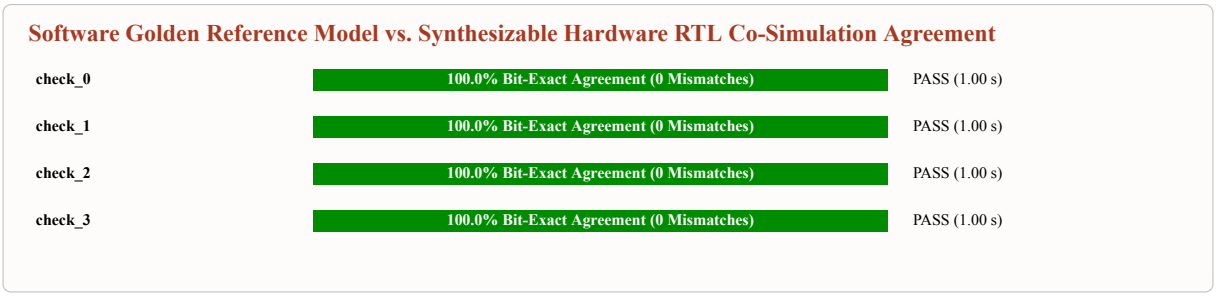


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.278 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

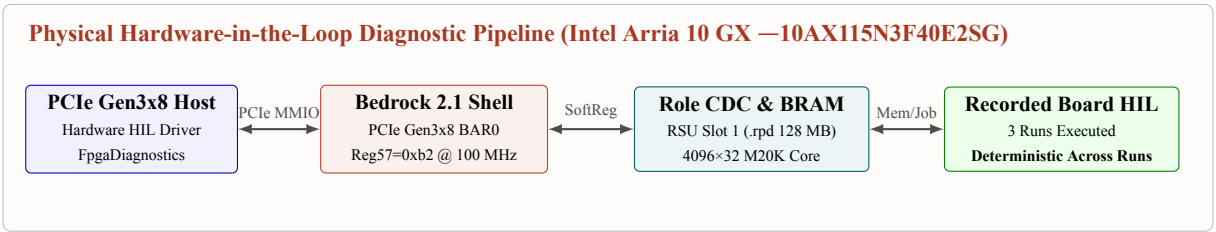


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Compile rtl/*.sv with hammer_v4_top as top; do not simultaneously compile duplicate module definitions from hammer_v3_reference/rtl/. Consult design/SPEC.md and design/ARCHITECTURE.md for detailed edge and maintenance contracts.

From the project root, run python3 software/build.py --out build/software --jobs 4 with a fresh destination. The builder refuses occupied or protected paths. The supplied Linux archive and patch hashes are recorded in software/source-manifest.json.

Required cross-compilers are /usr/bin/riscv64-unknown-elf-gcc 13.2.0 and /usr/bin/riscv64-linux-gnu-gcc 13.3.0 with binutils 2.42. Build dependencies include Python, make, host C/C++, dtc, cpio, tar, xz, bash, bc, bison, flex, perl and GNU patch; verification additionally requires Icarus, Verilator and Yosys.

Zero-initialize external platform RAM, then load images/fw_jump.bin at 0x80000000, images/Image at 0x80200000 and images/hammer-v4.dtb at 0x87e00000 using the generated image-manifest.json. Do not load vmlinux at its virtual ELF entry or inject /init separately.

Assert reset, release it with start low, wait for boot_ready without boot_error, then launch. Configure serial reception for 6,250,000 baud, 8N1. Verification supplies the line v4\n at frozen scenario times; premature RX can overrun before Linux enables reception.

Run python3 tests/verify.py for portable verification. Immutable per-seed result.json and uart-comparison.json files are under evidence/system-linux-validation-20261031T095914Z-ac456689; evidence/uart_linux_boot.log is the verified actual UART capture.

For physical integration, supply board-specific pin assignments and external interface timing constraints. The accepted standalone STA excludes input/output paths and must not be used as evidence of board-level interface timing closure.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128

W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 82 Verilog/SystemVerilog files; 15073 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
hammer_v3_reference/rtl/hammer_v2_alu.sv	34 total	34 code
hammer_v3_reference/rtl/hammer_v2_atomic.sv	228 total	228 code
hammer_v3_reference/rtl/hammer_v2_axi_lite.sv	94 total	94 code
hammer_v3_reference/rtl/hammer_v2_cache.sv	268 total	268 code
hammer_v3_reference/rtl/hammer_v2_clint.sv	89 total	89 code
hammer_v3_reference/rtl/hammer_v2_core.sv	864 total	861 code
hammer_v3_reference/rtl/hammer_v2_csr.sv	216 total	216 code
hammer_v3_reference/rtl/hammer_v2_dcache.sv	31 total	31 code
hammer_v3_reference/rtl/hammer_v2_decode.sv	228 total	228 code
hammer_v3_reference/rtl/hammer_v2_dmem.sv	28 total	28 code
hammer_v3_reference/rtl/hammer_v2_hazard.sv	24 total	24 code
hammer_v3_reference/rtl/hammer_v2_icache.sv	31 total	31 code
hammer_v3_reference/rtl/hammer_v2_imem.sv	28 total	28 code
hammer_v3_reference/rtl/hammer_v2_muldiv.sv	101 total	101 code
hammer_v3_reference/rtl/hammer_v2_predictor.sv	85 total	85 code
hammer_v3_reference/rtl/hammer_v2_ram.sv	136 total	136 code
hammer_v3_reference/rtl/hammer_v2_regfile.sv	23 total	23 code
hammer_v3_reference/rtl/hammer_v2_top.sv	299 total	297 code
hammer_v3_reference/rtl/hammer_v2_trap.sv	78 total	78 code
hammer_v3_reference/rtl/hammer_v3_accel_bridge.sv	248 total	245 code
hammer_v3_reference/rtl/hammer_v3_atomic.sv	229 total	229 code
hammer_v3_reference/rtl/hammer_v3_axi_geometry.sv	28 total	28 code
hammer_v3_reference/rtl/hammer_v3_axi_master.sv	142 total	141 code
hammer_v3_reference/rtl/hammer_v3_cache.sv	271 total	271 code
hammer_v3_reference/rtl/hammer_v3_cache_arbiter.sv	90 total	90 code
hammer_v3_reference/rtl/hammer_v3_core.sv	888 total	885 code
hammer_v3_reference/rtl/hammer_v3_csr.sv	223 total	223 code
hammer_v3_reference/rtl/hammer_v3_data_path.sv	121 total	121 code
hammer_v3_reference/rtl/hammer_v3_data_router.sv	101 total	101 code
hammer_v3_reference/rtl/hammer_v3_dcache.sv	31 total	31 code
hammer_v3_reference/rtl/hammer_v3_fabric.sv	148 total	148 code

Production file	Physical lines	Code lines
hammer_v3_reference/rtl/hammer_v3_fetch_path.sv	85 total	85 code
hammer_v3_reference/rtl/hammer_v3_icache.sv	31 total	31 code
hammer_v3_reference/rtl/hammer_v3_mmu.sv	299 total	296 code
hammer_v3_reference/rtl/hammer_v3_page_leaf.sv	38 total	38 code
hammer_v3_reference/rtl/hammer_v3_plic.sv	176 total	172 code
hammer_v3_reference/rtl/hammer_v3_ptw.sv	158 total	156 code
hammer_v3_reference/rtl/hammer_v3_tlb.sv	181 total	178 code
hammer_v3_reference/rtl/hammer_v3_top.sv	578 total	575 code
hammer_v3_reference/rtl/hammer_v3_uart.sv	418 total	411 code
rtl/hammer_v2_alu.sv	34 total	34 code
rtl/hammer_v2_atomic.sv	228 total	228 code
rtl/hammer_v2_axi_lite.sv	94 total	94 code
rtl/hammer_v2_cache.sv	268 total	268 code
rtl/hammer_v2_clint.sv	89 total	89 code
rtl/hammer_v2_core.sv	864 total	861 code
rtl/hammer_v2_csr.sv	216 total	216 code
rtl/hammer_v2_dcache.sv	31 total	31 code
rtl/hammer_v2_decode.sv	228 total	228 code
rtl/hammer_v2_dmem.sv	28 total	28 code
rtl/hammer_v2_hazard.sv	24 total	24 code
rtl/hammer_v2_icache.sv	31 total	31 code
rtl/hammer_v2_imem.sv	28 total	28 code
rtl/hammer_v2_muldiv.sv	101 total	101 code
rtl/hammer_v2_predictor.sv	85 total	85 code
rtl/hammer_v2_ram.sv	136 total	136 code
rtl/hammer_v2_regfile.sv	23 total	23 code
rtl/hammer_v2_top.sv	299 total	297 code
rtl/hammer_v2_trap.sv	78 total	78 code
rtl/hammer_v3_accel_bridge.sv	248 total	245 code
rtl/hammer_v3_atomic.sv	229 total	229 code
rtl/hammer_v3_axi_geometry.sv	28 total	28 code
rtl/hammer_v3_axi_master.sv	142 total	141 code
rtl/hammer_v3_cache.sv	271 total	271 code
rtl/hammer_v3_cache_arbiter.sv	90 total	90 code
rtl/hammer_v3_core.sv	888 total	885 code
rtl/hammer_v3_csr.sv	223 total	223 code
rtl/hammer_v3_data_path.sv	121 total	121 code
rtl/hammer_v3_data_router.sv	101 total	101 code
rtl/hammer_v3_dcache.sv	31 total	31 code
rtl/hammer_v3_fabric.sv	148 total	148 code
rtl/hammer_v3_fetch_path.sv	85 total	85 code

Production file	Physical lines	Code lines
rtl/hammer_v3_icache.sv	31 total	31 code
rtl/hammer_v3_mmu.sv	299 total	296 code
rtl/hammer_v3_page_leaf.sv	38 total	38 code
rtl/hammer_v3_plic.sv	176 total	172 code
rtl/hammer_v3_ptw.sv	158 total	156 code
rtl/hammer_v3_tlb.sv	181 total	178 code
rtl/hammer_v3_top.sv	578 total	575 code
rtl/hammer_v3_uart.sv	418 total	411 code
rtl/hammer_v4_boot_rom.sv	16 total	16 code
rtl/hammer_v4_top.sv	390 total	387 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E082	rtl/hammer_v3_top.sv
E101	tests/asset_checks.py
E103	tests/compatibility_checks.py
E113	tests/mutation_checks.py
E114	tests/parameters.py
E118	tests/reproduction_checks.py
E120	tests/rv64_reference.py
E121	tests/smode_checks.py
E122	tests/synthesis.py

Reference	Original project-relative record
E135	tests/system_oracle.py
E223	.vulcan/engineering/run-0001/attempt-0034/stdout.log
E225	.vulcan/engineering/run-0001/rtl-timing-lint.json
E226	.vulcan/engineering/run-0001/quartus-sta.json
E227	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).